

Set Theory for Computer Science

Week 6: Induction and Recursion

Mathematical induction

The well-ordering principle

Sequences and recursion

Transitive closure

The principle of mathematical induction

Mathematical induction is a strategy for proving claims of the form

S_n is true for every integer $n \geq a$

$$S_a$$

$$S_{a+1}$$

$$S_{a+2}$$

$$S_{a+3}$$

$$S_{a+4}$$

...

The principle of mathematical induction

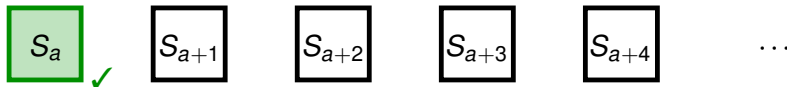
Mathematical induction is a strategy for proving claims of the form

S_n is true for every integer $n \geq a$

Strategy:

1. prove S_a is true

base case



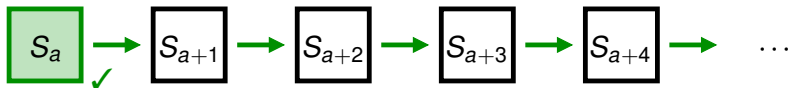
The principle of mathematical induction

Mathematical induction is a strategy for proving claims of the form

S_n is true for every integer $n \geq a$

Strategy:

1. prove S_a is true **base case**
2. prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$ **inductive step**



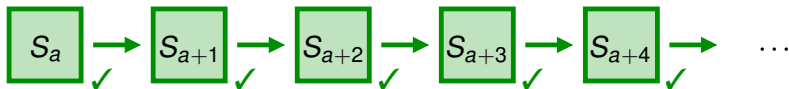
The principle of mathematical induction

Mathematical induction is a strategy for proving claims of the form

S_n is true for every integer $n \geq a$

Strategy:

1. prove S_a is true **base case**
2. prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$ **inductive step**



Formally:

$$S_a \wedge \forall m(m \geq a \rightarrow (S_m \rightarrow S_{m+1})) \rightarrow \forall n(n \geq a \rightarrow S_n)$$

The principle of mathematical induction

Mathematical induction is a strategy for proving claims of the form

S_n is true for every integer $n \geq a$

Strategy:

1. prove S_a is true **base case**
2. prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$ **induction step**
 - a) let $m \geq a$ be *arbitrary*
 - b) assume S_n is true for $n = m$ **Induction Hypothesis**
 - c) show this *implies* S_n for $n = m + 1$

$$S_a \wedge \forall m(m \geq a \rightarrow (S_m \rightarrow S_{m+1})) \rightarrow \forall n(n \geq a \rightarrow S_n)$$

Examples of mathematical induction

Claim for all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

For example, for $n = 4$, by binary addition:

$$\begin{array}{r} 2^0 \\ 2^1 \\ 2^2 \\ 2^3 \\ 2^4 \\ \hline 2^5 - 1 \end{array} + \begin{array}{r} 1 \\ 10 \\ 100 \\ 1000 \\ 10000 \\ \hline 11111 \end{array} +$$

Examples of mathematical induction

Claim for all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Proof – base case

► $\sum_{k=0}^0 2^k = 2^0 = 1$

► $2^{0+1} - 1 = 2^1 - 1 = 2 - 1 = 1$

Hence $\sum_{k=0}^n 2^k = 2^{n+1} - 1$ is true for $n = 0$



Examples of mathematical induction

Claim for all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Proof – induction step

Let $m \geq 0$. Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$

IH

Hence **IH** implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$



Examples of mathematical induction

Claim for all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Proof – induction step

Let $m \geq 0$. Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$

IH

$$\blacktriangleright \sum_{k=0}^{m+1} 2^k = \sum_{k=0}^m 2^k + 2^{m+1} \stackrel{\text{IH}}{=} 2^{m+1} - 1 + 2^{m+1}$$

$$\blacktriangleright 2^{(m+1)+1} - 1 = 2 \cdot 2^{m+1} - 1 = 2^{m+1} - 1 + 2^{m+1}$$

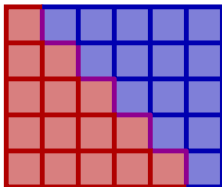
Hence **IH** implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$



Examples of mathematical induction

Claim for all $n \geq 1$, $\sum_{k=1}^n k = \frac{1}{2} n(n+1)$

For example, for $n = 5$:



$$\text{total area} = 2 \times (1 + 2 + 3 + 4 + 5) = 5 \times 6$$

Examples of mathematical induction

Claim for all $n \geq 1$, $\sum_{k=1}^n k = \frac{1}{2} n(n+1)$

Proof – base case

► $\sum_{k=1}^1 k = 1$

► $\frac{1}{2} \cdot 1 \cdot (1+1) = \frac{1}{2} \cdot 2 = 1$

Hence $\sum_{k=1}^n k = \frac{1}{2} n(n+1)$ is true for $n = 1$



Review exercise

Claim for all $n \geq 1$, $\sum_{k=1}^n k = \frac{1}{2} n(n+1)$

Exercise: complete the **induction step**

Let $m \geq 1$. Assume $\sum_{k=1}^m k = \frac{1}{2} m(m+1)$

IH

Hence **IH** implies $\sum_{k=1}^{m+1} k = \frac{1}{2} (m+1)((m+1)+1)$



Mathematical induction

The well-ordering principle

Sequences and recursion

Transitive closure

The well-ordering principle

Let $\leq$ be a partial order in B and $A \subseteq B$

An element $b \in B$ is a **lower bound for A** if $b \leq a$ for all $a \in A$

Examples

- ▶ 0 is a lower bound for $\{x: x > 0\}$ in $\mathbb{Z}$
- ▶ 0 is a lower bound for $\{x: x > 0\}$ in $\mathbb{Q}$
- ▶ 0 is a lower bound for $\{x: x > 0\}$ in $\mathbb{R}$

The well-ordering principle

Let $\leq$ be a partial order in B and $A \subseteq B$

An element $b \in B$ is a **lower bound for A** if $b \leq a$ for all $a \in A$

Examples

- ▶ 0 is a lower bound for $\{x: x > 0\}$ in $\mathbb{Z}$
- ▶ 0 is a lower bound for $\{x: x > 0\}$ in $\mathbb{Q}$
- ▶ 0 is a lower bound for $\{x: x > 0\}$ in $\mathbb{R}$

The well-ordering principle:

Every non-empty subset A of $\mathbb{Z}$ that has a lower bound has a smallest element (in the usual linear order on $\mathbb{Z}$)

Well-ordering implies mathematical induction

Theorem

In $\mathbb{Z}$, well-ordering implies mathematical induction

Well-ordering implies mathematical induction

Theorem

In $\mathbb{Z}$, well-ordering implies mathematical induction

Proof

Suppose

1. S_a is true
2. $\forall m(m \geq a \rightarrow (S_m \rightarrow S_{m+1}))$ is true

Define $A := \{n \in \mathbb{Z}: n \geq a \text{ and } S_n \text{ is false}\}$

Well-ordering implies mathematical induction

Theorem

In $\mathbb{Z}$, well-ordering implies mathematical induction

Proof

Suppose

1. S_a is true
2. $\forall m(m \geq a \rightarrow (S_m \rightarrow S_{m+1}))$ is true

Define $A := \{n \in \mathbb{Z} : n \geq a \text{ and } S_n \text{ is false}\}$



A cannot be non-empty! Hence S_n is true for all $n \geq a$

Mathematical induction implies well-ordering

Theorem

In $\mathbb{Z}$, mathematical induction implies well-ordering

Mathematical induction implies well-ordering

Theorem

In $\mathbb{Z}$, mathematical induction implies well-ordering

Proof

Suppose $A \subseteq \mathbb{Z}$ has lower bound b and no smallest element

We claim $\forall k (k < n \rightarrow k \notin A)$ is true for all $n \geq b$

Mathematical induction implies well-ordering

Theorem

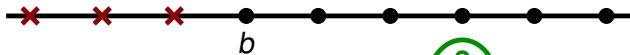
In $\mathbb{Z}$, mathematical induction implies well-ordering

Proof

Suppose $A \subseteq \mathbb{Z}$ has lower bound b and no smallest element

We claim $\forall k (k < n \rightarrow k \notin A)$ is true for all $n \geq b$

base case



induction



cannot be in A

Conclusion A must be empty!

break

Mathematical induction

The well-ordering principle

Sequences and recursion

Transitive closure

Sequences

A total function $f: \mathbb{N} \rightarrow A$ defines the **sequence**

$$(f(n))_{n=0}^{\infty} = (f(0), f(1), f(2), \dots)$$

- ▶ $f(n)$ is called the **n -th term** of the sequence
- ▶ we also use the notation $(t_n)_{n=a}^{\infty} = (t_a, t_{a+1}, t_{a+2}, \dots)$

Examples

- ▶ $f: \mathbb{N} \rightarrow \mathbb{N}$ with $f(n) = 2n \Rightarrow$ sequence $(0, 2, 4, 6, \dots)$
- ▶ $f: \mathbb{N} \rightarrow \mathbb{R}$ with $f(n) = \frac{1}{n+1} \Rightarrow$ sequence $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots)$

Recursively defined sequences

Sequences are often defined recursively

Example: geometric sequence

$(t_n)_{n=0}^{\infty}$ defined by $t_0 := c$, $t_{n+1} := t_n \cdot r$

Sequence $(c, c \cdot r, c \cdot r^2, c \cdot r^3, \dots)$ so $t_n = c \cdot r^n$

Example: factorials

$(t_n)_{n=0}^{\infty}$ defined by $t_0 := 1$, $t_{n+1} := (n + 1) \cdot t_n$

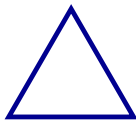
Sequence $(1, 1, 2 \cdot 1, 3 \cdot 2 \cdot 1, \dots)$ so $t_n = n!$

Recursively defined sequences

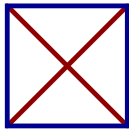
Example: counting diagonals

Let t_n be the number of diagonals of an n -gon. Then

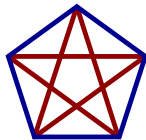
$$t_3 = 0, \quad t_{n+1} = t_n + (n - 1)$$



3-gon



4-gon



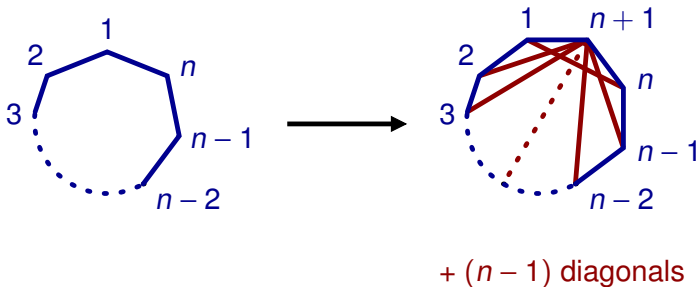
5-gon

Recursively defined sequences

Example: counting diagonals

Let t_n be the number of diagonals of an n -gon. Then

$$t_3 = 0, \quad t_{n+1} = t_n + (n - 1)$$



Number of diagonals of an n -gon

Claim the sequence $t_3 := 0$, $t_{n+1} := t_n + (n - 1)$ satisfies

$$t_n = \frac{1}{2}n(n - 3) \quad \text{for all } n \geq 3$$

Number of diagonals of an n -gon

Claim the sequence $t_3 := 0$, $t_{n+1} := t_n + (n - 1)$ satisfies

$$t_n = \frac{1}{2}n(n - 3) \quad \text{for all } n \geq 3$$

Proof – base case

- ▶ $t_3 = 0$
- ▶ $\frac{1}{2} \cdot 3 \cdot (3 - 3) = 0$

Hence $t_n = \frac{1}{2}n(n - 3)$ is true for $n = 3$



Review exercise

Claim the sequence $t_3 := 0$, $t_{n+1} := t_n + (n - 1)$ satisfies

$$t_n = \frac{1}{2}n(n - 3) \quad \text{for all } n \geq 3$$

Exercise: complete the **induction step**

Let $m \geq 3$. Assume $t_m = \frac{1}{2}m(m - 3)$

IH

Hence **IH** implies $t_{m+1} = \frac{1}{2}(m + 1)((m + 1) - 3)$



Mathematical induction

The well-ordering principle

Sequences and recursion

Transitive closure

Taking powers of a relation

For a binary relation R in a set A , define R^n ($n \geq 1$) recursively by

$$R^1 := R, \quad R^{n+1} := R^n \circ R$$

Then $R^1 = R, \quad R^2 = R \circ R, \quad R^3 = R \circ R \circ R, \quad \dots$

Interpretation of R^n

$x R^n y \iff$ there is a path of exactly n steps from x to y
in the directed graph representation of R

Transitive closure of a relation

The **transitive closure** of a binary relation R in a set A is the relation

$$R^* := \{\langle x, y \rangle : x R^n y \text{ for some } n \geq 1\}$$

Two equivalent definitions

1. $x R^* y \iff$ there is a path of finite length from x to y in the directed graph representation of R
2. R^* is the smallest transitive relation in A that contains R , i.e.

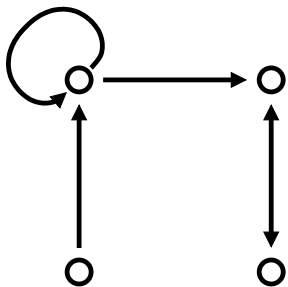
$$S \subseteq A \times A \text{ transitive and } R \subseteq S \Rightarrow R \subseteq R^* \subseteq S$$

Transitive closure of a relation

The **transitive closure** of a binary relation R in a set A is the relation

$$R^* := \{\langle x, y \rangle : x R^n y \text{ for some } n \geq 1\}$$

Example



Transitive closure of a relation

The **transitive closure** of a binary relation R in a set A is the relation

$$R^* := \{\langle x, y \rangle : x R^n y \text{ for some } n \geq 1\}$$

Example

